

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8a-jzb2-ms1c
Comments Due: March 15, 2025
Submission Type: API

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-2329
Comment on FR Doc # 2025-02305

Submitter Information

Email: [REDACTED]
Organization: SandboxAQ

General Comment

See attached file(s)

Attachments

SandboxAQ AI Action Plan Recommendations



FROM: SandboxAQ

SUBJECT: AI Action Plan

The United States and the Imperative of Artificial Intelligence (AI) Leadership

The United States has long benefited from the transformative power of artificial intelligence (AI), which has driven some of the most significant innovations of the past decade. However, our nation risks losing its competitive edge as peer competitors and adversaries rapidly advance beyond our current models and architectures. China, in particular, is aggressively developing AI-driven cyber capabilities and quantum decryption technologies, posing a direct threat to U.S. national security, economic leadership, and digital infrastructure.

While the U.S. has been at the forefront of AI innovation, concrete steps can be taken to safeguard our leadership and ensure that the benefits of AI and other advanced technologies continue to support American interests broadly.

Recognizing the vast potential of AI applications, it is essential to move beyond outdated misconceptions rooted in rigid organizational cultures and processes. AI has already demonstrated its ability to accelerate materials discovery, enhance cryptographic agility, and enable resilient navigation. To harness these capabilities fully, emerging technologies must be treated as strategic priorities. This urgency is particularly evident in the face of 'harvest now, decrypt later' threats, where adversaries collect encrypted U.S. data for future decryption using quantum technologies.

As Vice President J.D. Vance highlighted during the AI Action Summit, ensuring American AI leadership requires a comprehensive, cross-sector approach. The new AI Action Plan should implement strategic initiatives that unlock the nation's full competitive advantage in AI.

1. Secure U.S. AI Systems Against Quantum Cryptographic Attack.

Advances in quantum computing present a clear and urgent risk to AI-driven systems, military operations, financial networks, and digital infrastructure, making post-quantum cryptography (PQC) a national security priority.

Recommendations: The AI Action Plan should include provisions to accelerate and enhance the needed transition to PQC-secured AI systems across federal agencies and critical infrastructure. Among other things, the plan should call for the US government to: (1) implement PQC

standards across all federally funded AI programs, in alignment with NIST's post-quantum cryptographic guidelines; (2) establish mandatory cryptographic inventories across all federal agencies to assess quantum-vulnerable systems; (3) require agencies to deploy automated cryptography discovery tools to ensure continuous monitoring of cryptographic assets; and (4) mandate that all new federal AI systems be PQC-compatible, ensuring cryptographic agility from the outset.

2. Drive AI Innovation Across Key Technologies.

To achieve the Administration's vision for American AI dominance, disciplines adjacent to AI must mature in tandem with core AI research. An integrated approach will ensure the necessary infrastructures and capabilities are in place when the next technological revolution arrives. Rather than pursuing sequential development, we must recognize and capitalize on the documented benefits of AI across existing sectors and prepare to integrate and scale best-of-breed technologies at the nexus of AI and related disciplines.

Recommendations: The AI Action Plan should direct the Department of Defense (DoD) and the Department of Government Efficiency (DOGE) to conduct a comprehensive assessment of successful Other Transaction Authority (OTA) and Small Business Innovation Research (SBIR) programs that have demonstrated cross-sector potential. These assessments should lead to the conversion of promising initiatives into formal programs of record, thereby bridging the "valley of death" for nontraditional defense organizations and signaling robust economic investment across sectors. Priority areas should include alternative positioning, navigation, and timing (PNT) systems for contested logistics, network and security automation, post-quantum cryptography, simulation and optimization, and command and control (C2) automation.

3. Enhance Contract Authority Decision-Making through Market Intelligence.

As the AI market continues to evolve in quality and volume, contract officers and decision-makers must understand thoroughly the commercial environment that supports their portfolios. While each department has financial and technical operators, all acquisition personnel must demonstrate a rigorous grasp of AI and adjacent technology licensing models. The current "cost plus" pricing models are insufficient for scaling American AI dominance.

Recommendation: The AI Action Plan should direct the Office of Management and Budget (OMB) and the General Services Administration (GSA) to develop a new contracting framework—crafted in collaboration with industry stakeholders—that aligns AI-related contracts with contemporary commercial licensing practices. This approach will promote

transparency, drive cost efficiencies, and enable rapid scaling of AI capabilities to meet emerging national security challenges.

4. Integrate AI and Quantum Technologies.

The significant energy and computation demands associated with scaling AI systems present a clear challenge that advances in quantum computing can address. By investing in quantum technology as a critical enabler of AI, the United States can catalyze the innovation necessary to harness AI's potential.

Recommendation: The AI Action Plan should call for a National AI and Quantum Initiative dedicated to the joint study and maturation of quantum and AI technologies. This initiative would facilitate the integration of quantum-enhanced algorithms and secure AI architectures. Existing commercial technology pioneering work in merging quantum computing with AI security and efficiency serves as a model for a holistic approach that can drive a revolutionary transformation across multiple sectors.

5. Modernize Federal AI Procurement to Enable Rapid Adoption of Secure AI Solutions.

Federal procurement remains a significant bottleneck for AI adoption. Outdated acquisition models slow the deployment of critical AI security, quantum, and cryptographic resilience solutions.

Recommendation: The AI Action Plan should call for the federal government to (1) establish a government-wide AI Marketplace to allow pre-vetted AI vendors to deliver secure, FedRAMP-certified AI solutions rapidly; (2) expand OTA contracts for AI-driven cybersecurity and quantum-resistant applications; and (3) require cryptographic agility provisions in all federal AI contracts to ensure secure AI systems remain adaptable to emerging threats.

6. Foster Cross-Sector Collaboration and Innovation.

Recognizing that breakthroughs often occur at the nexus of disparate fields, promoting partnerships between government agencies, academia, and industry leaders is essential. Their advanced work in hybrid AI systems, secure computing, and quantum-enhanced algorithms underscores the importance of an integrated research and development ecosystem.

Recommendation: The AI Action Plan should direct key government agencies to establish a new wave of strategic partnerships and collaborative research initiatives that leverage the expertise

of innovative and disruptive organizations. By doing so, the United States can accelerate the translation of cross-disciplinary innovations into practical applications, ensuring sustained leadership in AI and related domains.

Conclusion

Securing America's leadership in AI requires a forward-looking, integrated approach that advances innovation across multiple disciplines. By fostering commercial development, streamlining contracting practices, integrating AI with quantum technologies, and enhancing cross-sector collaboration, the United States can strengthen its competitive edge in the evolving global AI landscape.

This revised AI Action Plan reflects a commitment to innovation, balancing established priorities with the transformative potential of today's commercial technology ecosystem. We must act decisively and collaboratively to remain at the forefront of the AI revolution.

Thank you for considering our recommendations. My team would happily discuss any of these matters with you further.

Regards,
Lloyd Dabbs
Director of Public Sector Business
SandboxAQ

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.